

Harsh Mangla

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EDUCATION

Chitkara University

Punjab, India

B.E. in Computer Science Engineering (AI & ML)

May 2024 – Expected 2028

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, DBMS, Statistics for Data Science

PROJECTS

Student Performance Prediction | Python, Scikit-learn, Pandas, Matplotlib

2025

- Built a multi-class classification pipeline using Logistic Regression, achieving **90%+ accuracy** on student grade prediction
- Conducted EDA to uncover key performance indicators; handled missing values and applied label/one-hot encoding
- Compared SVM, Decision Tree, and KNN; selected best model via cross-validation and hyperparameter tuning

Fraud Detection System | Python, Random Forest, SMOTE, Scikit-learn

2025

- Developed a fraud detection classifier using Random Forest on an imbalanced financial dataset with **99%+ AUC-ROC**
- Applied SMOTE oversampling to address class imbalance, improving minority class recall by **35%**
- Evaluated model robustness using confusion matrix, precision-recall curves, and k-fold cross-validation

Heart Disease Prediction | Python, Scikit-learn, Seaborn, Pandas

2025

- Engineered a binary classification model using Logistic Regression and SVM to predict cardiac risk with **88% accuracy**
- Performed feature importance analysis via correlation heatmaps; removed multicollinear features to boost performance
- Deployed end-to-end pipeline with train/test split, StandardScaler normalization, and confusion matrix evaluation

TECHNICAL SKILLS

Languages: Python, Java, SQL

ML & Data Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Microsoft Azure

Core Concepts: Machine Learning, Data Analysis, Feature Engineering, Model Evaluation, EDA, Data Preprocessing

CERTIFICATIONS

Microsoft Azure Data Fundamentals (DP-900) | Microsoft

2025

- Demonstrated knowledge of core data concepts, relational & non-relational data, analytics workloads, and Azure data services

Microsoft Azure Fundamentals (AZ-900) | Microsoft

2025

- Validated foundational understanding of cloud concepts, Azure architecture, pricing, and cloud service models (IaaS, PaaS, SaaS)

Microsoft Azure AI Fundamentals (AI-900) | Microsoft

2024

- Certified in AI/ML workloads on Azure including Computer Vision, NLP, Conversational AI, and Responsible AI principles

Programming in Python | Meta (Coursera)

2023

- Covered Python fundamentals: OOP, file I/O, functional programming, and data manipulation with lists/dicts

Introduction to Data Analytics | IBM (Coursera)

2023

- Learned data collection, wrangling, visualization, and statistical analysis using Python and Excel

Foundations: Data, Data, Everywhere | Google (Coursera)

2023

- Studied data-driven decision making, the data analysis lifecycle, spreadsheet tools, and basic SQL for data queries